Stochastic and Sequential Regimes: Extending the Decision Quotient to Dynamic Information Sufficiency

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February 23, 2026

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Zenodo DOI: 10.5281/zenodo.18140965

Abstract

We extend the static decision quotient framework of Paper 4 to stochastic and sequential regimes. Given a decision problem with actions A, factored state space $S = X_1 \times \cdots \times X_n$, and utility U, we ask: what complexity is required to identify sufficient coordinate sets when (a) the state is drawn from a distribution P, or (b) decisions unfold over time with transitions and observations?

Results:

- **Stochastic regime**: STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction. Tractable under product distributions, bounded support, log-concave.
- **Sequential regime**: SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction. Tractable under full observability, bounded horizon, tree structure.
- **Regime hierarchy**: Static $\subset$ Stochastic $\subset$ Sequential; coNP $\subset$ PP $\subset$ PSPACE.
- **Substrate cost**: The integrity-competence verdict is substrate-independent; trajectories diverge based on substrate cost κ.
- **Temporal learning**: Bayesian structure detection reduces abstention over time.
- **Temporal integrity**: Evidence-monotone retractions preserve integrity across sequences.

The reduction constructions are machine-checked in Lean 4 (3288 lines, 124 theorems).

Keywords: computational complexity, decision theory, stochastic decision problems, POMDPs, polynomial hierarchy, PSPACE

1 Introduction

Consider any decision problem $\mathcal{D}$ with actions A and factored state space $S = X_1 \times \cdots \times X_n$. A coordinate set $I \subseteq \{1, \ldots, n\}$ is *sufficient* if knowing only coordinates in I determines the optimal action:

$$s_I = s'_I \implies \mathrm{Opt}(s) = \mathrm{Opt}(s')$$

This is *coordinate sufficiency*—the formalization of *information sufficiency* (whether available information determines the optimal action) over a factored state space.

Paper 4 established that SUFFICIENCY-CHECK is coNP-complete for static decision problems (A, S, U). This paper extends that landscape to two natural extensions: stochastic and sequential regimes.

1.1 Stochastic Regime

When the state is drawn from a distribution $P : \Delta(S)$, sufficiency must account for randomness. We use the distribution-conditional definition: I is stochastically sufficient if $P(s \mid s_I) \otimes U$ determines the same optimal action as $P(s) \otimes U$. This subsumes expected-value sufficiency (where only the first moment matters) and connects to classical sufficient statistics.

Regime	Tuple	Complexity
Static	(A, S, U)	coNP-complete (Paper 4)
Stochastic	(A, S, U, P)	PP-complete
Sequential	(A, S, U, T, O)	PSPACE-complete

The PPupper bound comes from counting satisfying assignments weighted by P. The lower bound uses a MAJSAT reduction: given φ, construct uniform P over $\{0,1\}^n$ and let $U(\text{accept}, s) = \varphi(s)$, $U(\text{reject}, s) = 1/2$. Then $\emptyset$ is stochastically sufficient iff $\mathbb{E}_P[\varphi(s)] \geq 1/2$ or $< 1/2$—i.e., majority of assignments satisfy φ.

Tractable subcases: product distributions (expected utility decomposes), bounded support ($|\text{supp}(P)| \leq k$), log-concave distributions.

1.2 Sequential Regime

When decisions unfold over time with transitions $T : A \times S \to \Delta(S)$ and observations $O : S \to \Delta(\Omega)$, the problem becomes a POMDP. Sequential sufficiency asks whether observation history restricted to I-coordinates determines the optimal policy.

The PSPACEupper bound follows from Papadimitriou & Tsitsiklis (1987). The lower bound uses TQBF reduction: given $\forall x_1 \exists x_2 \forall x_3 \ldots \varphi$, construct n time steps, adversarial transitions at universal quantifiers, agent choices at existential quantifiers, terminal utility φ. Then full observation history is sequentially sufficient iff the QBF is true.

Tractable subcases: fully observable (MDP), bounded horizon, tree-structured transitions, deterministic transitions.

1.3 Substrate Cost

We extend Paper 4's integrity-competence matrix with substrate cost $\kappa(\text{cell}, \text{agentType})$. The matrix verdict is substrate-independent: $\text{verdict}(c)$ is the same for all agent types. But trajectories diverge: $\text{trajectory}(\sigma, \tau_1) \neq \text{trajectory}(\sigma, \tau_2)$ in general. This is the minimal formal hook for substrate in decision theory.

1.4 Temporal Learning

An agent facing sequential problem instances drawn from unknown distribution D can learn the structure class via Bayesian updates. As $k \to \infty$, posterior $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ if structure classes are identifiable. This reduces the abstention frontier: $A_1 \supseteq A_2 \supseteq \ldots$.

1.5 Temporal Integrity

Claims across sequences must satisfy evidence-monotone retractions: an agent retracts a claim only when new evidence contradicts it. Retraction without evidence violates temporal integrity; refinement ($I' \subset I$) is allowed.

1.6 Main Theorems

1. **Theorem 2.6:** STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction.

2. **Theorem 3.6:** SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction.

3. **Propositions 4.1 and 4.2 (Regime Hierarchy):** Static $\subset$ Stochastic $\subset$ Sequential; coNP$\subset$ PP$\subset$ PSPACE.

4. **Theorem 5.2 (Substrate Independence of Verdict):** For all cells c, verdict(c) is κ-independent.

5. **Theorem 5.3 (Substrate Dependence of Trajectory):** There exist τ_1, τ_2, σ with trajectory$(\sigma, \tau_1) \neq$ trajectory(σ, τ_2).

(L: CC58, CC48, CC64, CC63)

1.7 Claim-Typing Bridges (Inherited Static Core)

1. **Typed claim admissibility (inherited static core):** report admissibility is typed by certificate class (ABSTAIN, EXACT, EPSILON(ε)).

2. **Evidence-admissibility equivalence (inherited static core):** first-class evidence-object existence is equivalent to typed report admissibility.

3. **Empty-sufficient characterization (inherited static core):** the $I = \emptyset$ query is exactly universal constancy of the decision boundary, and non-sufficiency is exactly existence of a counterexample pair.

4. **Exact requires evidence (inherited static core):** exact admissibility is equivalent to exact evidence existence.

5. **No evidence implies zero certified confidence (inherited static core):** under signal consistency, no evidence implies zero certified confidence.

[C:P4-core typed-claim chain;R:DC,AR] (L: CC75,IC7,DP2,DP5,IC9,IC17)

1.8 Paper Structure

Section 2: Stochastic decision problems, PP-completeness, tractable cases. Section 3: Sequential (POMDP) problems, PSPACE-completeness, tractable cases. Section 4: Strict hierarchy of regimes. Section 5: Substrate cost formalization and independence results. Section 6: Bayesian structure detection, abstention reduction. Section 7: Temporal integrity constraints. Section 8: Transfer conditions between regimes. Section 9: Related work.

2 Stochastic Decision Problems

We extend the static decision problem to include a probability distribution over states.

2.1 Stochastic Decision Problem

Definition 2.1 (Stochastic Decision Problem)**.** A *stochastic decision problem with coordinate structure* is a tuple $\mathcal{D}_S = (A, X_1, \ldots, X_n, U, P)$ where:

- A is a finite set of *actions*
- $X_1, \ldots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \cdots \times X_n$ is the *state space*
- $U : A \times S \to \mathbb{Q}$ is the *utility function*
- $P \in \Delta(S)$ is a *distribution over states*

Definition 2.2 (Distribution-Conditional Sufficiency)**.** A coordinate set $I \subseteq \{1, \ldots, n\}$ is *stochastically sufficient* for $\mathcal{D}_S$ if:

$$P(s \mid s_I) \otimes U \text{ determines the same optimal action as } P(s) \otimes U.$$

Equivalently, for all $s, s' \in S$ with $s_I = s'_I$:

$$\arg\max_{a \in A} \mathbb{E}_{t \sim P(\cdot \mid s_I)}[U(a,t)] = \arg\max_{a \in A} \mathbb{E}_{t \sim P}[U(a,t)].$$

Definition 2.3 (Expected-Value Sufficiency (Alternative))**.** A coordinate set I is *expected-value sufficient* if:

$$\forall a, a' : \mathbb{E}_{s \sim P}[U(a,s) \mid s_I] \geq \mathbb{E}_{s \sim P}[U(a',s) \mid s_I] \implies a \in \mathrm{Opt}_P.$$

where $\mathrm{Opt}_P = \arg\max_{a \in A} \mathbb{E}_{s \sim P}[U(a,s)]$.

Decision 2.4. STOCHASTIC-SUFFICIENCY-CHECK: Given stochastic decision problem $\mathcal{D}_S = (A, S, U, P)$ and coordinate set I, is I stochastically sufficient?

Decision 2.5. STOCHASTIC-MINIMUM-SUFFICIENT-SET: Given $\mathcal{D}_S$, find the minimal $I \subseteq \{1, \ldots, n\}$ that is stochastically sufficient.

2.2 Complexity: PP-Completeness

Theorem 2.6 (STOCHASTIC-SUFFICIENCY-CHECK is PP-complete)**.** *STOCHASTIC-SUFFICIENCY-CHECK is PP-complete.*

1. **Membership in PP**: Given $\mathcal{D}_S$ and I, we must check whether knowing s_I determines the optimal action. This reduces to comparing expected utilities conditioned on s_I versus unconditioned. For each action pair (a, a'), we compare:

   $$\mathbb{E}_{s \sim P}[U(a,s) \mid s_I] \geq \mathbb{E}_{s \sim P}[U(a',s) \mid s_I].$$

   This requires counting weighted by P, which is in PP.

2. **PP-hardness via MAJSAT reduction**: Given Boolean formula φ on n variables, construct:

   - $S = \{0,1\}^n$ (state space = assignments)
   - P = uniform distribution over S
   - $A = \{\text{accept}, \text{reject}\}$

- $U(\text{accept}, s) = \varphi(s)$, $U(\text{reject}, s) = 1/2$
- $I = \emptyset$

Then $\emptyset$ is stochastically sufficient iff the optimal action is determined without seeing any coordinates:

$$\emptyset \text{ sufficient} \iff \mathbb{E}_{s\sim P}[\varphi(s)] \geq 1/2 \text{ or } < 1/2.$$

This is exactly MAJSAT, which is PP-complete.

(L: CC58, CC55)

Proof. For membership: We reduce to a PPpredicate. For each $a \in A$, compute $\mathbb{E}_P[U(a, s)]$ and compare. The comparison involves summing over $s \in S$ weighted by $P(s)$, which is polynomial in $|S|$ and the representation of P. The decision is whether the argmax is unique or whether knowing s_I changes it—this is a majority-vote comparison, hence in PP.

For hardness: The reduction from MAJSAT constructs the uniform distribution. If φ is true on $\geq 1/2$ of assignments, the expected utility of accepting is $\geq 1/2$, making accept optimal regardless of state observation. If $< 1/2$, reject is optimal. Thus $\emptyset$ is sufficient exactly when MAJSAT is true. ■

Example 2.7 (Stochastic Weather)**.** Consider stochastic umbrella decision:

- $A = \{\text{carry}, \text{don't carry}\}$
- $S = \{\text{rain}, \text{no rain}\}$
- $P(\text{rain}) = 0.3$, $P(\text{no rain}) = 0.7$
- $U(\text{carry}, \text{rain}) = 2$, $U(\text{carry}, \text{no rain}) = -1$
- $U(\text{don't carry}, \text{rain}) = -2$, $U(\text{don't carry}, \text{no rain}) = 0$

Expected utility: $E_P[U(\text{carry})] = 0.3(2)+0.7(-1) = -0.1$, $E_P[U(\text{don't carry})] = 0.3(-2)+0.7(0) = -0.6$. So carry is optimal. If we condition on forecast information, we check whether the conditional distribution changes the argmax.

2.3 Tractable Subcases

1. **Product distributions**: $P = P_1 \otimes \cdots \otimes P_n$. Then

$$\mathbb{E}_{s\sim P}[U(a, s) \mid s_I] = \prod_{i\in I} P_i(s_i) \cdot \mathbb{E}_{s\sim P}[U(a, s) \mid s_I],$$

   so expected utility decomposes. This reduces stochastic sufficiency to static sufficiency per coordinate.

2. **Bounded support**: $|\text{supp}(P)| \leq k$. Enumeration is $O(k \cdot |A|)$, polynomial.

3. **Log-concave distributions**: For ordered state spaces, conditional expectations are monotone and computable via convex optimization.

Claim 2.8 (Tractability under product distributions)**.** If $P = P_1 \otimes \cdots \otimes P_n$ is a product distribution, then STOCHASTIC-SUFFICIENCY-CHECK reduces to checking static sufficiency on each marginal. *[C:P2.8;R:PD]*

(L: CC60)

Claim 2.9 (Tractability under bounded support)**.** If $|\text{supp}(P)| \leq k$, then STOCHASTIC-SUFFICIENCY-CHECK is solvable in $O(k \cdot |A| \cdot n)$ time. *[C:P2.9;R:BS]*

(L: CC59)

2.4 Bridge from Static

Paper 4 proved that static sufficiency does NOT imply stochastic sufficiency (CC54). A counterexample exists where I is sufficient for (A, S, U) but not for (A, S, U, P).

However, under product distributions, static sufficiency transfers:

Claim 2.10 (Static to stochastic transfer)**.** If $P = P_1 \otimes \cdots \otimes P_n$ is a product distribution and I is statically sufficient for (A, S, U), then I is stochastically sufficient for (A, S, U, P). *[C:P2.10;R:PD]*

(L: CC52)

3 Sequential Decision Problems

We further extend to sequential decision-making with transitions and observations.

3.1 Sequential Decision Problem (POMDP)

Definition 3.1 (Sequential Decision Problem)**.** A *sequential decision problem with coordinate structure* is a tuple $\mathcal{D}_{SEQ} = (A, X_1, \ldots, X_n, U, T, O)$ where:

- A is a finite set of *actions*
- $X_1, \ldots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \cdots \times X_n$ is the *state space*
- $U : A \times S \to \mathbb{Q}$ is the *stage utility*
- $T : A \times S \to \Delta(S)$ is the *transition kernel*
- $O : S \to \Delta(\Omega)$ is the *observation model*

The agent chooses a policy $\pi : (\Omega^*) \to A$ mapping observation histories to actions. Objective: maximize expected cumulative utility $\mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t U(a_t, s_t)\right]$ with discount factor $\gamma \in [0, 1)$.

Notation 3.2 (Observation History)**.** Let $\omega_t = (o_0, o_1, \ldots, o_t)$ denote the observation history up to time t. The agent's policy is $\pi(\omega_{t-1}) = a_t$.

Definition 3.3 (Sequential Sufficiency)**.** A coordinate set $I \subseteq \{1, \ldots, n\}$ is *sequentially sufficient* for $\mathcal{D}_{SEQ}$ if the optimal policy based on full observation history is the same as the optimal policy based only on coordinates in I at each time step.

Formally: let π^* be the optimal policy with full observations, and let π_I^* be the optimal policy with observations restricted to I-coordinates. Then $\pi^* = \pi_I^*$ (as functions of observation history).

Decision 3.4. SEQUENTIAL-SUFFICIENCY-CHECK: Given sequential decision problem $\mathcal{D}_{SEQ}$ and coordinate set I, is I sequentially sufficient?

Decision 3.5. SEQUENTIAL-MINIMUM-SUFFICIENT-SET: Given $\mathcal{D}_{SEQ}$, find the minimal I that is sequentially sufficient.

3.2 Complexity: PSPACE-Completeness

Theorem 3.6 (SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete)**.** *SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete.*

1. **Membership in PSPACE**: Optimal POMDP policy computation is in PSPACE(Papadimitriou & Tsitsiklis, 1987). Checking whether a coordinate set I yields the same policy involves comparing value functions, which can be done in polynomial space via alternating polynomial time.

2. **PSPACE-hardness via TQBF reduction**: Given quantified Boolean formula $\forall x_1 \exists x_2 \forall x_3 \ldots \varphi(x_1, \ldots, x_n)$, construct:

   - n time steps, one variable per step
   - At odd steps (universal quantifier): "nature" chooses the variable (adversarial transition)
   - At even steps (existential quantifier): agent chooses the variable (action)
   - Terminal utility: $\varphi(x_1, \ldots, x_n)$
   - Observation: agent sees all previous variables
   - I = full observation history

   Then full observation is sequentially sufficient iff the agent can guarantee $\varphi = \text{true}$ regardless of nature's choices, which holds iff the QBF is true.

(L: CC48, CC47)

Construction 3.7 (TQBF Reduction)**.** Given QBF $\Phi = Q_1 x_1 Q_2 x_2 \ldots Q_n x_n \ \varphi(x_1, \ldots, x_n)$ with alternating quantifiers, construct sequential problem:

- Time steps $t = 1, \ldots, n$
- State at step t encodes $(x_1, \ldots, x_t)$
- At step t: - If $Q_t = \forall$: transition is adversarial (nature chooses both next state and observation) - If $Q_t = \exists$: transition is controlled by agent action (choose x_t)
- Terminal reward: 1 if $\varphi(x_1, \ldots, x_n) = \text{true}$, 0 otherwise
- Full observation reveals all previous variables

Then: full observation is sequentially sufficient $\iff$ there exists a policy guaranteeing $\varphi = \text{true}$ $\iff$ Φ is true.

(L: CC77)

Claim 3.8 (Sequential sufficiency implies static sufficiency)**.** When the horizon $T = 1$ and transitions are deterministic, sequential sufficiency reduces to static sufficiency. *[C:P3.8;R:T=1,DET]*

(L: CC46)

3.3 Tractable Subcases

1. **Fully observable (MDP)**: O is the identity (agent sees full state). Then the problem reduces to a stochastic (or deterministic) problem per time step. Value iteration is polynomial in $|S|, |A|, T$.

2. **Bounded horizon**: $T \leq k$. The state space grows as $O(|S|^T)$, but for fixed T this is polynomial in $|S|$.

3. **Tree-structured transitions**: The transition graph has no cycles. Dynamic programming yields polynomial algorithms.

4. **Deterministic transitions**: $T(s'|a, s)$ is a point mass. The problem reduces to search over deterministic paths.

Claim 3.9 (Tractability under full observability)**.** If O is the identity (fully observable), then SEQUENTIAL-SUFFICIENCY-CHECK reduces to checking stochastic sufficiency at each time step. *[C:P3.9;R:FO]*

(L: CC50)

Claim 3.10 (Tractability under bounded horizon)**.** If $T \leq k$ (constant horizon), then SEQUENTIAL-SUFFICIENCY-CHECK is solvable in time polynomial in $|S|, |A|, k$. *[C:P3.10;R:BH]*

(L: CC49)

3.4 Bridge from Static and Stochastic

Paper 4 proved that static sufficiency does NOT transfer to sequential (horizon > 1): CC18

Similarly, stochastic sufficiency does NOT transfer to sequential in general:

Claim 3.11 (Stochastic to sequential bridge failure)**.** There exists a stochastic sufficient coordinate set I and a sequential problem where I is NOT sequentially sufficient, even when transitions are memoryless. *[C:P3.11;R:CE]*

(L: CC56)

The transfer conditions are:

- Static $\rightarrow$ Sequential: transfers iff $T = 1$ and transitions are deterministic (one-step bridge from Paper 4)

- Stochastic $\rightarrow$ Sequential: transfers iff transitions are memoryless AND observation model is regime-compatible

4 Regime Hierarchy

We establish the strict inclusion hierarchy among decision regimes.

Inclusion	Complexity Classes	Condition
Static $\subset$ Stochastic	coNP$\subset$ PP	Standard assumptions (P $\neq$ coNP)
Stochastic $\subset$ Sequential	PP$\subset$ PSPACE	Standard assumptions (P $\neq$ PP)

Claim 4.1 (Strict inclusion static $\subset$ stochastic)**.** Under standard complexity assumptions (P $\neq$ coNP), there exist stochastic decision problems that are not statically reducible. *[C:P4.1;R:P $\neq$ coNP]*

(L: P4SS1.)

Claim 4.2 (Strict inclusion stochastic $\subset$ sequential)**.** Under standard complexity assumptions (P $\neq$ PP), there exist sequential decision problems that are not stochastically reducible. *[C:P4.2;R:P $\neq$ PP]*

(L: P4SS2.)

4.1 Integrity-Resource Bound per Regime

The integrity-resource bound from Paper 4 generalizes to each regime:

1. **Static** (coNP-hard): polynomial-time agent must abstain on some instances
2. **Stochastic** (PP-hard): polynomial-time agent must abstain on MORE instances
3. **Sequential** (PSPACE-hard): polynomial-time agent must abstain on EVEN MORE instances

Formally, if $C_1 \subsetneq C_2$ (complexity classes), then the set of instances where integrity forces abstention under C_1-resources is a strict superset of those under C_2-resources:

$$\{\sigma : \text{integrity forces abstention at } C_1\} \supsetneq \{\sigma : \text{integrity forces abstention at } C_2\}$$

Claim 4.3 (Abstention frontier shifts)**.** As regime complexity increases (static $\to$ stochastic $\to$ sequential), the abstention frontier expands. *[C:P4.3;R:RG]*

(L: CC39)

4.2 Regime Detection

A meta-problem: given a problem instance, which regime is it in?

Decision 4.4. REGIME-DETECTION: Given a problem instance, is it static, stochastic, or sequential?

Conjecture: regime detection is polynomial for the represented family:

- Bounded actions $\to$ static
- Distribution present $\to$ stochastic
- Transitions/observations present $\to$ sequential

(L: CC41)

5 Substrate Cost

We extend the integrity-competence matrix from Paper 4 with substrate cost κ.

5.1 Definition

Definition 5.1 (Substrate Cost Function)**.** $\kappa : (\text{MatrixCell} \times \text{AgentType}) \to (\mathbb{R}_{\geq 0}, \leq)$

Where:

- MatrixCell = (integrity, attempted, competenceAvailable) triple from Paper 4
- AgentType = opaque type representing substrate (silicon, carbon, formal system, etc.)
- The codomain is a totally ordered cost space

Claim 5.2 (Verdict is substrate-independent)**.** For all cells c and agent types τ_1, τ_2:

$$\text{verdict}(c; \tau_1) = \text{verdict}(c; \tau_2).$$

[C:T5.2;R:RG]

(L: CC64)

Claim 5.3 (Trajectory is substrate-dependent)**.** There exist agent types τ_1, τ_2 and problem sequences σ such that:

$$\text{trajectory}(\sigma; \tau_1) \neq \text{trajectory}(\sigma; \tau_2)$$

even though $\text{verdict}(\sigma_t; \tau_1) = \text{verdict}(\sigma_t; \tau_2)$ for all t. *[C:T5.3;R:RG]*

(L: CC63)

5.2 What κ Captures

κ is a placeholder for everything the (A, S, U) tuple doesn't model:

- Memory
- Motivation
- Fatigue
- Identity
- Embodiment

We do NOT decompose κ. We acknowledge it exists, show where it enters the dynamics, and prove that the matrix verdict is κ-independent while the trajectory is κ-dependent.

Claim 5.4 (Minimal formal hook)**.** $\kappa(\text{cell}, \text{agentType})$ is the sufficient statistic of substrate for the trajectory problem. *[C:T5.4;R:RG]*

(L: CC65)

This is the clean separation: the model captures exactly the behaviorally relevant component of substrate, and nothing more.

We do NOT prove that κ captures all of subjective experience. We prove that κ captures all of subjective experience that is *decision-relevant*—i.e., that affects the agent's trajectory through the problem sequence.

5.3 Where Substrate Enters

The matrix verdict is computed from (integrity, attempted, competenceAvailable). None of these depend on substrate. The verdict is universal.

Substrate enters at exactly one point: the COST of being in a cell. This cost determines:

- Whether the agent attempts the next instance or rests
- How the agent allocates resources across a sequence
- The agent's effective learning rate (some substrates learn faster)
- The agent's abstention threshold (some substrates tolerate uncertainty better)

This connects to Paper 5 (Credibility): $\kappa = \kappa_{\text{compute}} + \kappa_{\text{communicate}}$, where $\kappa_{\text{communicate}}$ is bounded by Paper 5's cheap talk framework.

6 Temporal Learning

We formalize how an agent learns structure classes across sequential problem instances.

6.1 Setup

Agent faces a sequence of problem instances $I_1, I_2, \ldots$ drawn from an unknown distribution D over problem structures.

After solving instance I_k (or abstaining), the agent observes:

- Whether I_k belonged to a tractable subcase
- Which structure class C_k was detected (if any)
- The cost paid (substrate-dependent κ)

Claim 6.1 (Bayesian structure detection)**.** Agent maintains posterior $P(C \mid I_1, \ldots, I_k)$ over structure classes. Update rule: standard Bayesian update with likelihood $P(I_k \mid C)$. *[C:T6.1;R:RG]*

(L: CC4)

Convergence: does $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ as $k \to \infty$?

Answer: yes, if the structure classes are identifiable (distinct likelihood functions).

Claim 6.2 (Identifiability implies convergence)**.** If structure classes C are identifiable (distinct likelihood functions), then $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ as $k \to \infty$. *[C:P6.2;R:ID]*

(L: CC19)

6.2 PAC Learning Connection

Sample complexity of structure detection: How many instances must the agent solve before it can identify the structure class with probability $\geq 1 - \delta$ and error $\leq \varepsilon$?

Claim 6.3 (Sample complexity conjecture)**.** For the represented family (bounded actions, separable utility, tree structure), sample complexity is polynomial in $1/\varepsilon$ and $\log(1/\delta)$. *[C:J6.3;R:RG]*

This connects to PAC learning theory—the structure classes are the hypothesis space.

6.3 Learning Reduces Abstention

As the agent learns the structure class, it can apply regime-appropriate heuristics. The abstention frontier shrinks over time—not because the problem gets easier, but because the agent's model of the problem gets more accurate.

Formally: let A_k be the set of instances where integrity forces abstention at time k. Then:

$$A_1 \supseteq A_2 \supseteq \ldots \supseteq A_\infty$$

where A_∞ = instances genuinely outside all tractable subcases.

Claim 6.4 (Abstention reduction over time)**.** A_k is decreasing in k: $A_1 \supseteq A_2 \supseteq \ldots$. *[C:T6.4;R:RG]*

(L: CC68)

The agent's trajectory diverges based on κ even though verdicts are κ-independent. Different substrates learn at different rates, have different abstention thresholds, and make different trade-offs between solving and abstaining.

This connects to Paper 5: temporal learning produces costly signals. Each solved instance is a costly signal in Paper 5's sense: producing a correct solution has truth-dependent cost. The agent's track record is costly signal; self-assessment is cheap talk.

7 Temporal Integrity

We formalize consistency of claims across sequences of problem instances.

7.1 Definition

Definition 7.1 (Temporal Integrity)**.** Temporal integrity = consistency of claims across a sequence of instances.

An agent has temporal integrity if:

- Each per-instance claim satisfies Paper 4's integrity constraint
- Retractions are evidence-monotone: the agent retracts a claim only when new evidence contradicts it, never without cause
- Refinements are monotone: if the agent claims I is sufficient at time t, and later claims $I' \subset I$ is sufficient at time $t' > t$, this is a refinement (allowed). If it claims $I' \supset I$, this is a weakening that requires justification.

Decision 7.2. TEMPORAL-INTEGRITY-CHECK: Given a sequence of claims and evidence, is the sequence temporally integrity-preserving?

Claim 7.3 (Retraction with evidence preserves integrity)**.** If an agent claimed I sufficient, observed a counterexample, then retracted, this preserves temporal integrity. *[C:P7.3;R:RG]*

(L: CC70)

Claim 7.4 (Retraction without evidence violates integrity)**.** If an agent claimed I sufficient, observed no new information, then retracted, this violates temporal integrity. *[C:P7.4;R:RG]*

(L: CC71)

Claim 7.5 (Refinement strengthens integrity)**.** If an agent claimed I sufficient, discovers $I' \subset I$ also sufficient, updates claim to I'. This preserves and strengthens temporal integrity. *[C:P7.5;R:RG]*

(L: CC69)

7.2 Complexity of Temporal Integrity

Claim 7.6 (Complexity lower bound)**.** TEMPORAL-INTEGRITY-CHECK is at least as hard as the base regime's SUFFICIENCY-CHECK. *[C:T7.6;R:RG]*

Each claim in the sequence requires verification. The temporal constraint (monotonicity) adds structure but doesn't reduce worst-case complexity.

Paper 4 proves integrity forces abstention. Paper 5 proves claiming integrity is cheap talk. Together: the agent must abstain (Paper 4), cannot credibly communicate why (Paper 5, cheap talk bound), and can only escape by producing costly signals (Paper 5, Theorem 4.1)—which are exactly the mechanized proofs that Paper 4 itself provides.

8 Cross-Regime Transfer

When does a sufficiency result in regime R_1 help in regime R_2?

8.1 Transfer Conditions

1. **Static → Stochastic**: Transfers iff distribution P concentrates on a tractable static subcase. Formally: if $\text{supp}(P)$ lies within a structure-detectable class C, then static sufficiency for C implies stochastic sufficiency under P. Otherwise: bridge failure (already mechanized).

2. **Static → Sequential**: Transfers iff horizon $T = 1$ and transitions are deterministic. This is the one-step bridge (already proved in Paper 4: `CC30`). For $T > 1$: bridge failure (`CC18`).

3. **Stochastic → Sequential**: Transfers iff transitions are memoryless AND observation model is regime-compatible. Formally: if $T(s'|a, s)$ depends only on the stochastically sufficient coordinates of s, then stochastic sufficiency lifts to sequential sufficiency. Otherwise: new bridge-failure witness needed.

4. **Sequential → Static**: Always degenerates. A sequential problem with $T = 1$ and deterministic transitions IS a static problem.

Claim 8.1 (Transfer static to stochastic)**.** Static sufficiency transfers to stochastic iff P concentrates on tractable subcase. *[C:T8.1;R:RG]*

(L: `CC53`)

Claim 8.2 (Transfer stochastic to sequential)**.** Stochastic sufficiency transfers to sequential iff memoryless transitions and regime-compatible observations. *[C:T8.2;R:RG]*

(L: `CC57`)

8.2 Heuristic Transfer Across Regimes

Paper 4's heuristic reusability bridge generalizes: if structure detection is polynomial in regime R_1, and the detected structure is preserved under regime extension to R_2, then the heuristic transfers.

Key question: when is structure preserved?

- Product distributions preserve separable utility
- Deterministic transitions preserve tree structure

9 Related Work

This paper extends the static decision quotient to stochastic and sequential regimes. Related work falls into several categories.

9.1 Complexity of Dynamic Decision Problems

The complexity of MDPs and POMDPs is well-established: optimal MDP planning is in P, optimal POMDP planning is PSPACE-complete [10, 8, 6]. Our contribution extends this to the meta-question of identifying sufficient coordinates in these settings.

Sufficient statistics [3] is the classical foundation. Our stochastic sufficiency definition connects to this tradition but focuses on decision-relevant information rather than statistical estimation.

9.2 Information and Signaling

Value of information [5, 11] quantifies payment for information. We address a different question: the complexity of identifying which information is sufficient for decision.

Signaling games [12, 9, 7] connect to our temporal integrity and credibility results (Paper 5). The agent must abstain (Paper 4), cannot credibly signal why (Paper 5), and can only escape via costly signals.

9.3 Feature Selection in Machine Learning

Feature selection asks which inputs predict outputs, known NP-hard in general [2, 1, 4]. Our decision-theoretic analog for sufficiency is PP-complete (stochastic) and PSPACE-complete (sequential).

9.4 Regime Complexity Hierarchy

Our hierarchy Static $\subset$ Stochastic $\subset$ Sequential with complexity classes coNP $\subset$ PP $\subset$ PSPACE extends the representation-sensitive complexity results established for MDPs/POMDPs to the meta-level sufficiency problem.

This paper integrates: (i) stochastic and sequential extensions, (ii) substrate cost formalization, (iii) temporal learning and integrity, and (iv) cross-regime transfer conditions.

10 Conclusion

This paper extends the static decision quotient to stochastic and sequential regimes.

Summary of Results

- **Stochastic sufficiency** (Section 2): STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction. Tractable under product distributions, bounded support, log-concave.
- **Sequential sufficiency** (Section 3): SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction. Tractable under full observability, bounded horizon, tree structure.
- **Regime hierarchy** (Section 4): Static $\subset$ Stochastic $\subset$ Sequential; coNP $\subset$ PP $\subset$ PSPACE.

- **Substrate cost** (Section 5): Verdict is κ-independent; trajectory is κ-dependent.
- **Temporal learning** (Section 6): Bayesian structure detection reduces abstention over time.
- **Temporal integrity** (Section 7): Evidence-monotone retractions preserve integrity.

These results place stochastic and sequential sufficiency at higher complexity levels than static sufficiency.

Open Questions

- **Exact transfer conditions:** Characterize when stochastic sufficiency transfers to sequential.
- **PAC learning bounds:** Prove sample complexity bounds for structure detection.
- **Temporal integrity complexity:** Tight complexity bounds for TEMPORAL-INTEGRITY-CHECK.

This work extends Paper 4's framework to dynamic settings while preserving the core insight: identifying sufficient information is computationally hard, and integrity forces abstention when resources don't match problem complexity.

11 Lean 4 Proof Listings

The complete Lean 4 formalization extends the Paper 4 artifact. The mechanization consists of 3288 lines across 29 files, with 124 theorem/lemma statements.
Handle IDs. Inline theorem metadata cites compact IDs (for example, `CC4`, `HD2`, `IC3`) instead of full theorem constants.

11.1 Lean Handle ID Map

ID	Full Lean handle
CC1	DecisionQuotient.ClaimClosure.RegimeSimulation
CC2	DecisionQuotient.ClaimClosure.adq_ordering
CC3	DecisionQuotient.ClaimClosure.agent_transfer_licensed_iff_snapshot
CC4	DecisionQuotient.ClaimClosure.anchor_sigma2p_complete_conditional
CC5	DecisionQuotient.ClaimClosure.anchor_sigma2p_reduction_core
CC6	DecisionQuotient.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset
CC7	DecisionQuotient.ClaimClosure.bounded_actions_detectable
CC8	DecisionQuotient.ClaimClosure.bridge_boundary_represented_family
CC9	DecisionQuotient.ClaimClosure.bridge_failure_witness_non_one_step
CC10	DecisionQuotient.ClaimClosure.bridge_transfer_iff_one_step_class
CC11	DecisionQuotient.ClaimClosure.certified_total_bits_split_core
CC12	DecisionQuotient.ClaimClosure.cost_asymmetry_eth_conditional
CC13	DecisionQuotient.ClaimClosure.declaredBudgetSlice
CC14	DecisionQuotient.ClaimClosure.declaredRegimeFamily_complete
CC15	DecisionQuotient.ClaimClosure.declared_physics_no_universal_exact_certifier_core

ID	Full Lean handle (continued)
CC16	DecisionQuotient.ClaimClosure.dichotomy_conditional
CC17	DecisionQuotient.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core
CC18	DecisionQuotient.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core
CC19	DecisionQuotient.ClaimClosure.exact_certainty_inflation_under_hardness_core
CC20	DecisionQuotient.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core
CC21	DecisionQuotient.ClaimClosure.exact_raw_only_of_no_exact_admissible_core
CC22	DecisionQuotient.ClaimClosure.explicit_assumptions_required_of_not_excused_core
CC23	DecisionQuotient.ClaimClosure.explicit_state_upper_core
CC24	DecisionQuotient.ClaimClosure.hard_family_all_coords_core
CC25	DecisionQuotient.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
CC26	DecisionQuotient.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
CC27	DecisionQuotient.ClaimClosure.information_barrier_opt_oracle_core
CC28	DecisionQuotient.ClaimClosure.information_barrier_state_batch_core
CC29	DecisionQuotient.ClaimClosure.information_barrier_value_entry_core
CC30	DecisionQuotient.ClaimClosure.integrity_resource_bound_for_sufficiency
CC31	DecisionQuotient.ClaimClosure.integrity_universal_applicability_core
CC32	DecisionQuotient.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice
CC33	DecisionQuotient.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice
CC34	ComplexityClass.lt
CC35	DecisionQuotient.ClaimClosure.minsuff_collapse_to_conp_conditional
CC36	DecisionQuotient.ClaimClosure.minsuff_conp_complete_conditional
CC37	DecisionQuotient.ClaimClosure.no_auto_minimize_of_p_neq_conp
CC38	DecisionQuotient.ClaimClosure.no_exact_claim_admissible_under_hardness_core
CC39	DecisionQuotient.ClaimClosure.no_exact_claim_under_declared_assumptions_unless_excused_core
CC40	DecisionQuotient.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized
CC41	DecisionQuotient.ClaimClosure.no_uncertified_exact_claim_core
CC42	DecisionQuotient.ClaimClosure.one_step_bridge
CC43	DecisionQuotient.ClaimClosure.oracle_lattice_transfer_as_regime_simulation
CC44	DecisionQuotient.ClaimClosure.physical_crossover_above_cap_core
CC45	DecisionQuotient.ClaimClosure.physical_crossover_core
CC46	DecisionQuotient.ClaimClosure.physical_crossover_hardness_core
CC47	DecisionQuotient.ClaimClosure.physical_crossover_policy_core
CC48	DecisionQuotient.ClaimClosure.process_bridge_failure_witness
CC49	DecisionQuotient.ClaimClosure.query_obstruction_boolean_corollary
CC50	DecisionQuotient.ClaimClosure.query_obstruction_finite_state_core
CC51	DecisionQuotient.ClaimClosure.regime_core_claim_proved
CC52	DecisionQuotient.ClaimClosure.regime_simulation_transfers_hardness
CC53	DecisionQuotient.ClaimClosure.reusable_heuristic_of_detectable
CC54	DecisionQuotient.ClaimClosure.selectorSufficient_not_implies_setSufficient
CC55	DecisionQuotient.ClaimClosure.separable_detectable
CC56	DecisionQuotient.ClaimClosure.snapshot_vs_process_typed_boundary
CC57	DecisionQuotient.ClaimClosure.standard_assumption_ledger_unpack

ID	Full Lean handle (continued)
CC58	DecisionQuotient.ClaimClosure.stochastic_objective_bridge_can_fail_on_sufficiency
CC59	DecisionQuotient.ClaimClosure.subproblem_hardness_lifts_to_full
CC60	DecisionQuotient.ClaimClosure.subproblem_transfer_as_regime_simulation
CC61	DecisionQuotient.ClaimClosure.sufficiency_conp_complete_conditional
CC62	DecisionQuotient.ClaimClosure.sufficiency_conp_reduction_core
CC63	DecisionQuotient.ClaimClosure.sufficiency_iff_dq_ratio
CC64	DecisionQuotient.ClaimClosure.sufficiency_iff_projectedOptCover_eq_opt
CC65	DecisionQuotient.ClaimClosure.thermo_conservation_additive_core
CC66	DecisionQuotient.ClaimClosure.thermo_energy_carbon_lift_core
CC67	DecisionQuotient.ClaimClosure.thermo_eventual_lift_core
CC68	DecisionQuotient.ClaimClosure.thermo_hardness_bundle_core
CC69	DecisionQuotient.ClaimClosure.thermo_mandatory_cost_core
CC70	DecisionQuotient.ClaimClosure.tractable_bounded_core
CC71	DecisionQuotient.ClaimClosure.tractable_separable_core
CC72	DecisionQuotient.ClaimClosure.tractable_subcases_conditional
CC73	DecisionQuotient.ClaimClosure.tractable_tree_core
CC74	DecisionQuotient.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency
CC75	DecisionQuotient.ClaimClosure.tree_structure_detectable
CC76	DecisionQuotient.ClaimClosure.typed_claim_admissibility_core
CC77	DecisionQuotient.ClaimClosure.typed_static_class_completeness
CC78	DecisionQuotient.ClaimClosure.universal_solver_framing_core
CR1	DecisionQuotient.ConfigReduction.config_sufficiency_iff_behavior_preserving
DP1	DecisionQuotient.DecisionProblem.minimalSufficient_iff_relevant
DP2	DecisionQuotient.DecisionProblem.relevantSet_is_minimal
DP3	DecisionQuotient.DecisionProblem.sufficient_implies_selectorSufficient
DP4	DecisionQuotient.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
DP5	DecisionQuotient.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient
DP6	DecisionProblem.anchorSufficient
DP7	DecisionProblem.classMonotoneOn
DP8	DecisionProblem.constant_opt_all_sufficient
DQ1	DQ.ClaimClosure.tractable_tree_core
DQ2	DQ.ClaimClosure.tree_structure_detectable
DQ3	DQ.ClaimClosure.typed_claim_admissibility_core
DQ4	DQ.ClaimClosure.typed_static_class_completeness
DQ6	DQ.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
DQ7	DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
DQ8	DQ.Sigma2PHardness.representationGap_eq_zero_iff
HD1	DecisionQuotient.HardnessDistribution.centralization_dominance_bundle
HD2	DecisionQuotient.HardnessDistribution.centralization_step_saves_n_minus_one
HD3	DecisionQuotient.HardnessDistribution.centralized_higher_leverage
HD4	DecisionQuotient.HardnessDistribution.complete_model_dominates_after_threshold
HD5	DecisionQuotient.HardnessDistribution.gap_conservation_card

ID	Full Lean handle (continued)
HD6	DecisionQuotient.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant
HD7	DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness
HD8	DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
HD9	DecisionQuotient.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
HD10	DecisionQuotient.HardnessDistribution.generalized_right_eventually_dominates_wrong
HD11	DecisionQuotient.HardnessDistribution.hardnessEfficiency_eq_central_share
HD12	DecisionQuotient.HardnessDistribution.isRightHardness
HD13	DecisionQuotient.HardnessDistribution.isWrongHardness
HD14	DecisionQuotient.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
HD15	DecisionQuotient.HardnessDistribution.native_dominates_manual
HD16	DecisionQuotient.HardnessDistribution.no_positive_slope_linear_represents_saturating
HD17	DecisionQuotient.HardnessDistribution.right_dominates_wrong
HD18	DecisionQuotient.HardnessDistribution.saturatingSiteCost_eventually_constant
HD19	DecisionQuotient.HardnessDistribution.simplicityTax_grows
HD20	DecisionQuotient.HardnessDistribution.totalDOF_eventually_constant_iff_zero_distributed
HD21	DecisionQuotient.HardnessDistribution.totalDOF_ge_intrinsic
HD22	DecisionQuotient.HardnessDistribution.hardnessLowerBound
IC1	DecisionQuotient.IntegrityCompetence.CertaintyInflation
IC2	DecisionQuotient.IntegrityCompetence.CompletionFractionDefined
IC3	DecisionQuotient.IntegrityCompetence.EvidenceForReport
IC4	DecisionQuotient.IntegrityCompetence.ExactCertaintyInflation
IC5	DecisionQuotient.IntegrityCompetence.Percent
IC6	DecisionQuotient.IntegrityCompetence.RLFFWeights
IC7	DecisionQuotient.IntegrityCompetence.ReportSignal
IC8	DecisionQuotient.IntegrityCompetence.ReportBitModel
IC9	DecisionQuotient.IntegrityCompetence.SignalConsistent
IC10	DecisionQuotient.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
IC11	DecisionQuotient.IntegrityCompetence.admissible_matrix_counts
IC12	DecisionQuotient.IntegrityCompetence.abstain_signal_exists_with_guess_self
IC13	DecisionQuotient.IntegrityCompetence.certaintyInflation_iff_not_admissible
IC14	DecisionQuotient.IntegrityCompetence.certificationOverheadBits
IC15	DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_evidence
IC16	DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_no_evidence
IC17	DecisionQuotient.IntegrityCompetence.certifiedTotalBits
IC18	DecisionQuotient.IntegrityCompetence.certifiedTotalBits_ge_raw
L1	ExactPhysicalClaimWellTyped
L2	ExcusedBy
L3	ExplicitHardnessAssumptions
L4	Literal.eval
L5	OneStepSequentialObjective
L6	RegimeSimulation
L7	StandardComplexityAssumptions

ID	Full Lean handle (continued)
L8	StochasticCriterionObjective
L9	StructureDetectable
L10	TransitionCoupledObjective
L11	TwoStepObjective
L12	adq
L13	adq_ordering
L14	agentBridgeClass
L15	agent_transfer_licensed_iff_snapshot
L16	anchor_sigma2p_complete_conditional
L17	anchor_sigma2p_reduction_core
L18	boundaryCharacterized
L19	boundaryCharacterized_iff_exists_sufficient_subset
L20	bounded_actions_detectable
L21	bounded_actions_reusable_heuristic
L22	bridgeFailureWitness
P4SS1	Paper4bStochasticSequential.ClaimClosure.claim_static_stochastic_product
P4SS2	Paper4bStochasticSequential.ClaimClosure.claim_stochastic_sequential_memoryless
PBC1	DecisionQuotient.PhysicalBudgetCrossover.CrossoverAt
PBC2	DecisionQuotient.PhysicalBudgetCrossover.explicit_infeasible_succinct_feasible_of_crossover
PBC3	DecisionQuotient.PhysicalBudgetCrossover.has_crossover_of_bounded_succinct_unbounded_explicit

11.2 What Is Machine-Checked

The Lean formalization establishes:

1. **STOCHASTIC-SUFFICIENCY PP-completeness**: Reduction from MAJSAT showing stochastic sufficiency checking is PP-complete.
2. **SEQUENTIAL-SUFFICIENCY PSPACE-completeness**: Reduction from TQBF showing sequential sufficiency checking is PSPACE-complete.
3. **Substrate independence theorems**: Formal proofs that the matrix verdict is κ-independent while trajectories are κ-dependent.
4. **Bridge transfer conditions**: Formalized conditions under which sufficiency transfers between regimes.

11.3 Assumption Ledger (Auto)

Source files.

- `Paper4bStochasticSequential/ClaimClosure.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

11.4 Complete Claim Coverage Matrix

Paper handle	Status	Lean support
cor:exact-identifiability	Derived	DQ.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
cor:exact-no-competence-zero-certified	Full	DQ.IntegrityCompetence.certificationOverheadBits_of_no_evidence
cor:gap-externalization	Full	DQ.HardnessDistribution.hardnessLowerBound, DQ.HardnessDistribution.simplicityTax_grows
cor:gap-minimization-hard	Derived	DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
cor:generalized-eventual-dominance	Full	DQ.HardnessDistribution.generalized_right_eventually_dominates_wrong
cor:information-barrier-query	Full	DQ.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core, DQ.ClaimClosure.exact_raw_only_of_no_exact_admissible_core, DQ.ClaimClosure.explicit_assumptions_required_of_not_excused_core
cor:integrity-universal	Full	DQ.ClaimClosure.hard_family_all_coords_core
cor:linear-positive-no-saturation	Full	DQ.HardnessDistribution.no_positive_slope_linear_represents_saturating
cor:no-auto-minimize	Full	DQ.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
cor:no-uncertified-exact-claim	Full	DQ.ClaimClosure.information_barrier_value_entry_core
cor:outside-excuses-no-exact-report	Full	DQ.ClaimClosure.information_barrier_opt_oracle_core
cor:overmodel-diagnostic-implication	Derived	DQ.Sigma2PHardness.representationGap_eq_zero_iff
cor:physics-no-universal-exact-claim	Full	DQ.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
cor:practice-bounded	Full	agentBridgeClass
cor:practice-diagnostic	Derived	DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
cor:practice-structured	Full	agent_transfer_licensed_iff_snapshot
cor:practice-tree	Full	anchor_sigma2p_complete_conditional
cor:practice-unstructured	Full	ExactPhysicalClaimWellTyped
cor:query-obstruction-bool	Full	DQ.ClaimClosure.no_auto_minimize_of_p_neq_conp, OneStepSequentialObjective
cor:right-wrong-hardness	Full	DQ.HardnessDistribution.right_dominates_wrong
cor:type-system-threshold	Full	DQ.HardnessDistribution.native_dominates_manual
prop:abstention-frontier	Full	DQ.ClaimClosure.no_exact_claim_under_declared_assumptions_unless_excused_core
prop:adq-ordering	Full	DQ.ClaimClosure.adq_ordering
prop:attempted-competence-matrix	Full	DQ.IntegrityCompetence.EvidenceForReport, DQ.IntegrityCompetence.ExactCertaintyInflation, DQ.IntegrityCompetence.certaintyInflation_iff_not_admissible
prop:bridge-failure-horizon	Full	DQ.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core, DQ.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core
prop:bridge-failure-stochastic	Full	DQ.ClaimClosure.selectorSufficient_not_implies_setSufficient
prop:bridge-failure-transition	Derived	DQ.ClaimClosure.tractable_tree_core
prop:bridge-transfer-scope	Full	DQ.ClaimClosure.integrity_resource_bound_for_sufficiency
prop:budgeted-crossover	Full	DQ.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice, DQ.PhysicalBudgetCrossover.explicit_infeasible_succinct_feasible_of_crossover
prop:certified-confidence-gate	Full	DQ.IntegrityCompetence.certificationOverheadBits, DQ.IntegrityCompetence.certificationOverheadBits_of_evidence
prop:crossover-above-cap	Full	DQ.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice, DQ.PhysicalBudgetCrossover.has_crossover_of_bounded_succinct_unbounded_explicit
prop:crossover-not-certification	Full	CC.lt

prop:crossover-policy	Full	DQ.ClaimClosure.minsuff_collapse_to_conp_conditional
prop:declared-contract-selection-validity	Full	DQ.ClaimClosure.declaredRegimeFamily_complete, DQ.ClaimClosure.information_barrier_opt_oracle_core
prop:dominance-modes	Full	DQ.HardnessDistribution.centralized_higher_leverage
prop:empty-sufficient-constant	Full	DQ.DecisionProblem.relevantSet_is_minimal
prop:evidence-admissibility-equivalence	Full	DQ.IntegrityCompetence.ReportSignal
prop:exact-requires-evidence	Full	DQ.IntegrityCompetence.ReportBitModel, DQ.IntegrityCompetence.SignalConsistent
prop:generalized-assumption-boundary	Full	DQ.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness, DQ.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
prop:hardness-conservation	Full	DQ.HardnessDistribution.totalDOF_ge_intrinsic
prop:hardness-efficiency-interpretation	Full	DQ.HardnessDistribution.hardnessEfficiency_eq_central_share
prop:heuristic-reusability	Full	DQ.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset, DQ.ClaimClosure.oracle_lattice_transfer_as_regime_simulation, DQ.ClaimClosure.physical_crossover_core
prop:identifiability-convergence	Full	DQ.ClaimClosure.exact_certainty_inflation_under_hardness_core
prop:insufficiency-counterexample	Full	DQ.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient, DQ.DecisionProblem.minimalSufficient_iff_relevant
prop:integrity-competence-separation	Full	DQ.IntegrityCompetence.Percent, DQ.IntegrityCompetence.admissible_matrix_counts
prop:integrity-resource-bound	Full	DQ.ClaimClosure.explicit_state_upper_core, DQ.IntegrityCompetence.abstain_signal_exists_with_guess_self, DQ.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
prop:mdp-tractable	Full	DQ.ClaimClosure.query_obstruction_finite_state_core
prop:minimal-relevant-equiv	Full	DP.anchorSufficient, DQ.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
prop:no-evidence-zero-certified	Full	DQ.IntegrityCompetence.certifiedTotalBits
prop:one-step-bridge	Full	DQ.ClaimClosure.integrity_resource_bound_for_sufficiency
prop:oracle-lattice-strict	Full	ExcusedBy, ExplicitHardnessAssumptions
prop:oracle-lattice-transfer	Full	DQ.ClaimClosure.integrity_universal_applicability_core, boundaryCharacterized
prop:outside-excuses-explicit-assumptions	Full	DQ.ClaimClosure.declaredRegimeFamily_complete
prop:physics-no-universal-exact	Full	DQ.ClaimClosure.cost_asymmetry_eth_conditional
prop:query-finite-state-generalization	Full	RegimeSimulation, adq
prop:query-randomized-robustness	Full	L.eval, TransitionCoupledObjective, TwoStepObjective
prop:query-randomized-weighted	Full	bounded_actions_reusable_heuristic, bridgeFailureWitness
prop:query-regime-obstruction	Full	DQ.ClaimClosure.no_exact_claim_admissible_under_hardness_core, RegimeSimulation, adq
prop:query-state-batch-lb	Full	StandardComplexityAssumptions, adq_ordering
prop:query-subproblem-transfer	Full	DQ.ClaimClosure.one_step_bridge, DQ.ClaimClosure.sufficiency_conp_complete_conditional, DQ.ClaimClosure.sufficiency_conp_reduction_core
prop:query-tightness-full-scan	Full	StructureDetectable
prop:query-value-entry-lb	Full	StochasticCriterionObjective, anchor_sigma2p_reduction_core
prop:query-weighted-transfer	Full	boundaryCharacterized_iff_exists_sufficient_subset, bounded_actions_detectable
prop:refinement-strengthens	Full	DQ.ClaimClosure.thermo_mandatory_cost_core
prop:retraction-evidence-integrity	Full	DQ.ClaimClosure.tractable_bounded_core
prop:retraction-no-evidence-violates	Full	DQ.ClaimClosure.tractable_separable_core
prop:selector-separation	Full	DQ.ClaimClosure.physical_crossover_above_cap_core

prop: sequential-bounded-horizon	Full	DQ.ClaimClosure.query_obstruction_boolean_corollary
prop: sequential-static-relation	Full	DQ.ClaimClosure.physical_crossover_hardness_core
prop:set-to-selector	Full	DP.constant_opt_all_sufficient
prop: snapshot-process-typing	Full	DQ.ClaimClosure.agent_transfer_licensed_iff_snapshot, DQ.ClaimClosure.minsuff_conp_complete_conditional, DQ.ClaimClosure.regime_core_claim_proved
prop: static-stochastic-strict	Full	P4SS.ClaimClosure.claim_static_stochastic_product
prop: static-stochastic-transfer	Full	DQ.ClaimClosure.regime_simulation_transfers_hardness
prop: stochastic-bounded-support	Full	DQ.ClaimClosure.subproblem_hardness_lifts_to_full
prop:stochastic-product-tractable	Full	DQ.ClaimClosure.subproblem_transfer_as_regime_simulation
prop:stochastic-sequential-bridge-fail	Full	DQ.ClaimClosure.snapshot_vs_process_typed_boundary
prop:stochastic-sequential-strict	Full	P4SS.ClaimClosure.claim_stochastic_sequential_memoryless
prop:sufficiency-char	Full	DQ.ClaimClosure.thermo_energy_carbon_lift_core, DQ.ClaimClosure.thermo_eventual_lift_core
prop:thermo-conservation-additive	Full	DQ.ClaimClosure.tractable_subcases_conditional
prop: thermo-hardness-bundle	Full	DQ.ClaimClosure.tree_structure_detectable
prop:thermo-lift	Full	DQ.ClaimClosure.tractable_tree_core, DQ.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency
prop:thermo-mandatory-cost	Full	DQ.ClaimClosure.typed_claim_admissibility_core
prop: typed-claim-admissibility	Derived	DQ.ClaimClosure.tree_structure_detectable
prop: universal-solver-framing	Derived	DQ.ClaimClosure.typed_static_class_completeness
prop: zero-epsilon-competence	Full	DQ.IntegrityCompetence.RLFFWeights, DQ.IntegrityCompetence.certifiedTotalBits_ge_raw
prop: zero-epsilon-reduction	Full	DP.classMonotoneOn, DQ.DecisionProblem.sufficient_implies_selectorSufficient
thm:amortization	Full	DQ.HardnessDistribution.complete_model_dominates_after_threshold
thm:bridge-boundary-represented	Full	DQ.ClaimClosure.bounded_actions_detectable, DQ.ClaimClosure.bridge_boundary_represented_family, DQ.ClaimClosure.bridge_failure_witness_non_one_step
thm: centralization-dominance	Full	DQ.HardnessDistribution.centralization_dominance_bundle, DQ.HardnessDistribution.centralization_step_saves_n_minus_one
thm:claim-integrity-meta	Full	DQ.ClaimClosure.declaredRegimeFamily_complete
thm:config-reduction	Full	DQ.ConfigReduction.config_sufficiency_iff_behavior_preserving
thm:cost-asymmetry-eth	Full	DQ.ClaimClosure.bridge_transfer_iff_one_step_class, DQ.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
thm:dichotomy	Full	DQ.ClaimClosure.declaredBudgetSlice, DQ.ClaimClosure.declared_physics_no_universal_exact_certifier_core, DQ.ClaimClosure.dichotomy_conditional
thm:generalized-dominance	Full	DQ.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
thm:generalized-saturation-possible	Full	DQ.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant, DQ.HardnessDistribution.saturatingSiteCost_eventually_constant
thm: linear-saturation-iff-zero	Full	DQ.HardnessDistribution.totalDOF_eventually_constant_iff_zero_distributed
thm:overmodel-diagnostic	Full	DQ.ClaimClosure.anchor_sigma2p_reduction_core, DQ.ClaimClosure.information_barrier_state_batch_core
thm:regime-coverage	Full	DQ.ClaimClosure.certified_total_bits_split_core, DQ.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized
thm:tax-conservation	Full	DQ.HardnessDistribution.gap_conservation_card
thm:tax-grows	Full	DQ.HardnessDistribution.hardnessLowerBound
thm:tractable	Full	DQ.ClaimClosure.universal_solver_framing_core
thm: typed-completeness-static	Derived	DQ.ClaimClosure.typed_claim_admissibility_core

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found.
Auto summary: mapped 102/102 (full=94, derived=8, unmapped=0).

11.5 Claims Not Fully Mechanized

Entries marked **Full (conditional)** are explicitly mechanized transfer theorems that depend on standard external complexity facts (e.g., MAJSAT PP-completeness, TQBF PSPACE-completeness), with those dependencies represented as hypotheses in Lean.

11.6 Verification

The proofs compile with Lean 4. Run `lake build` in the proof directory to verify.

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